EIT-CDAE: A 2-D Electrical Impedance Tomography Image Reconstruction Method Based on Auto Encoder Technique

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***Abstract*—Electrical Impedance Tomography is considered to be an alternative substitution to CT and MRI technologies as it is a non-invasive, safe medical imaging technology, and free of ionizing or heating radiation. Similar to CT and MRI technologies, reconstructing a two-dimensional EIT image is also considered an ill-posed and non-linear inverse problem, where the image quality is highly sensitive to the measurement data, and often random noise artifacts appear in the image with the different non-linear algorithms. Therefore, in this work, we have proposed a new EIT image reconstruction algorithm based on the convolution denoising autoencoder (CDAE) deep learning algorithm. Our EIT-CDAE used a convolutional neural network in the encoder and decoder network. From our experimental data using phantom data, our EIT-CDAE model has reconstructed a better EIT image quality, removing any noise artifacts, making it more robust compared to the conventional stacked autoencoder and traditional non-linear algorithms. The source code is available in the github: https://github.com/yongfu-li/eit-cdae-algorithm**

***Index Terms*—Electrical impedance tomography, image reconstruction, autoencoder, convolutional neural network, deep learning**

I. Introduction

Electrical Impedance Tomography (EIT) is a clinical imaging technology [1], where its operating principle involves injecting a small and safe amount of current into the human body and measuring the induced voltages using multiple electrodes [2]. Hence, EIT is considered a non-invasive, safe medical imaging technology and is free of ionizing or heating radiation. Its advantages of non-invasive, radiation-free and low cost compared to computed tomography (CT) and magnetic resonance imaging (MRI) have brought broad prospects for EIT. It is also reported as the only viable wearable real-time method for lung imaging [3]–[5] at present.

As shown in Fig. 1, an EIT system consists of a hardware measurement system, an image reconstruction software algorithm, which is suitable for continuous regional lung monitoring, preventing mechanical pulmonary injury [4]–[6]. Our 16-electrode EIT hardware system injects an alternating current into the region between two adjacent electrodes, and voltage signals are measured on all other electrodes in turn, i.e.

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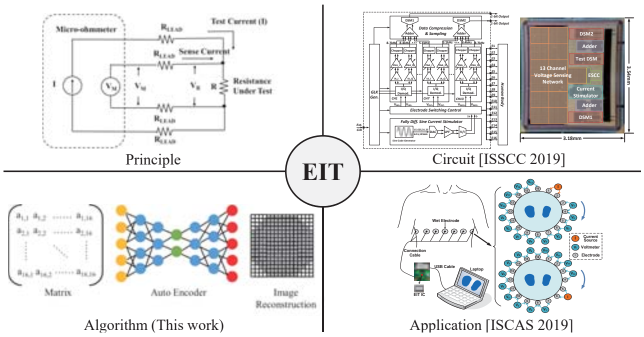

Fig. 1. Overview of EIT system

13 channels for voltage recording across the chest. Sequential pairs are then successively used for injecting current until all possible combinations have been measured. Finally, a complete data set of 208 (13×16) combinations is collected. The measurements could be utilized to recover one frame tomographic image of the lung.

Our current system [5], [6] uses Graz consensus Reconstruction algorithm for EIT(GREIT) [7] , implemented in the Electrical Impedance Tomography and Diffuse Optical Tomography Reconstruction Software(EIDORS) platform [8]. EIDORS is an open source software toolbox written mainly in MATLAB/OCTAVE, which is designed for solving the forward and inverse modeling problems [8]. The toolbox is divided into three stages i.e. mesh generation, forward problem, and inverse problem. The forward problem calculates the voltage values inside the object when conductivity distribution is given in that object. The inverse problem is to obtain an adequate estimation of the interior conductivity distribution, based on the measured voltage values. The inverse problem is also considered as a seriously ill-posed problem, where any arbitrarily small perturbations in the measured data can produce arbitrarily large perturbations of the solution, causing the reconstructed image to be extremely sensitive to modeling errors and measurement. In addition, the number of measurements is much less than that of the unknown conductivity

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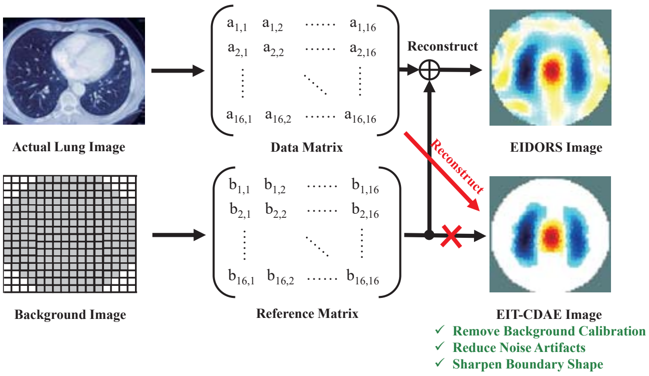

Fig. 2. Comparison of EIT reconstructed images using the (a) EIDORS and (b) our proposed work EIT-CDAE.

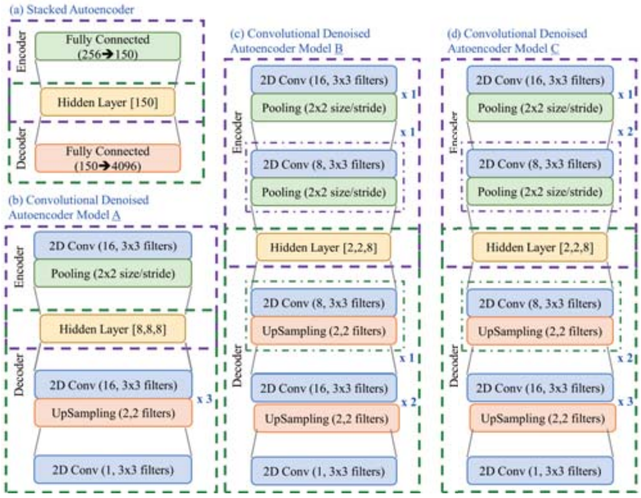

Fig. 3. Different type of autoencoder architectures. (a) stacked autoencoder (SAE) (b)-(d) our proposed work EIT-CDAE with the different numbers of convolutional layers, pooling and upsampling layers.

parameters to be reconstructed, leading to an under-determined problem.

As shown in Fig. 2, GREIT method [7] uses a differential imaging method, where the images is reconstructed based on the conductivity between EIT measurements at two times instants. Often, an EIT system is evaluated using a saline phantom by computing the difference between the background data (homogeneous medium in an acrylic tank) and actual measurement data (conductive and insulting objects in acrylic tank). Hence, the difference EIT method requires a stable and accurate background measurement/calibration, which is impractical in real-life application scenarios [7], [9], [10].

Thanks to the escalated development in deep learning technology, there have been gratifying breakthroughs in classification, processing and recognition fields for image, audio, video, speech, and natural languages. Researchers began to adopt deep learning technology to solve the ill-posed inverse problems in the EIT image reconstruction and have made considerable achievements [11]–[14]. A basic convolutional neural network (CNN) has been proposed to post process the EIT image and improve the image contrast, leading to sharp and reliable reconstructions even for the highly nonlinear inverse problems [15], [16]. The stacked autoencoder (SAE) is also an effective approach to enhancing the image quality [13]. Nevertheless, both algorithms are sensitivity to measurement, resulting in random artifacts that appears in the images.

The contributions of our work are summarized as follows:

- We proposed a new EIT convolutional denoising autoencoder (CDAE) model, combining convolutional layers in the autoencoder model, to perform two-dimensional EIT image reconstruction algorithm. Our model does not background calibration compared to the GREIT method, reduces noise artifacts and sharpen the boundary shape of any arbitrary object in the tank.
- We conducted a performance analysis on EIT-CDAE to identify an optimal neural network required for our application.
- An open source database [17] are used to evaluate our proposed EIT-CDAE algorithm.

The rest of the paper is organized as follows. Section II details our proposed deep learning model, EIT-CDAE and section III presents the experimental results of our proposed EIT-CDAE models and the performance analysis using different combinations of convolutional layers and fully connected layers. Finally, conclusions are drawn in Section IV.

II. Deep Learning Models for EIT Image Reconstruction

A. Autoencoder Architecture

As shown in Fig. 3(a), an autoencoder is a neural network that has two basic networks and a hidden layer: an input network (encoder), a hidden layer, and an output network (decoder). This network is a machine learning architecture that is trained to map its inputs to a different type of output. This network has several distinct advantages over different neural networks such as multilayer perceptron network (MLP), convolutional neural network (CNN) and recurrent neural network (RNN). For example, autoencoder has been used to perform colorization of a gray-scale image [18] and to reconstruct the input from a corrupted version of it [19]. Recently, it has been demonstrated to solve an ill-posed linear inverse data recovery problem [11]. This motivated us to look further in this architecture to solve our EIT image reconstruction problem.

B. Our Proposed Architecture: EIT-CDAE

In this work, we have proposed a convolutional denoising auto-encoder (CDAE) network for our EIT image reconstruction problem. The architecture of our EIT-CDAE is illustrated in Fig. 3(b)-(d). Other than the state-of-the-art SAE architecture (Fig. 3(a)) used in [13], [20], convolutional layers in the encoder and decoder networks is introduced to reduce memory usage and improve the image quality. Hence, the encoder and decoder networks consist of multiple two-dimensional convolutional layers and fully connected layers. The two-dimensional convolutional layers consist of different levels

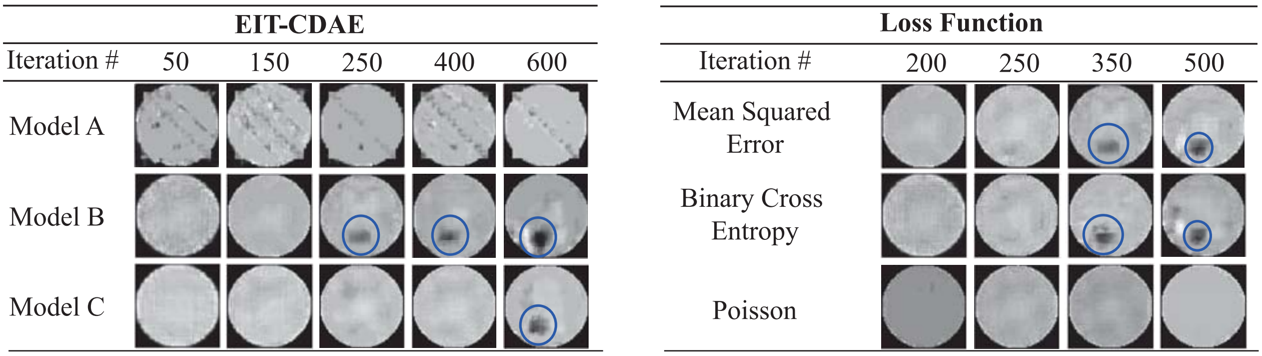

Fig. 4. (a)Reconstructed images using our proposed EIT-CDAE and (b) the use of different loss function to evaluate the model B.

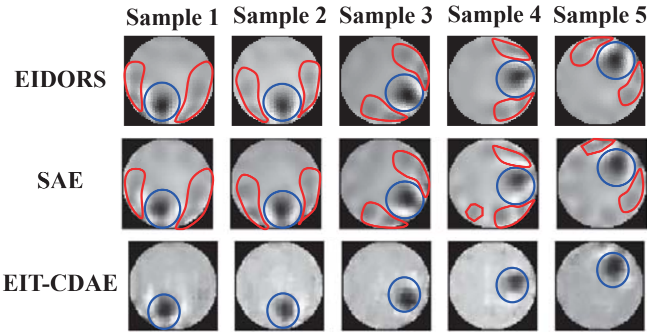

Fig. 5. Reconstructed images from (a) EIDORS (b) Stacked Autoencoder (SAE) (c) EIT-CDAE.

of depth (1, 8 and 16) with the same kernel size of 3×3. We have selected the rectified linear units (ReLU) as the activation function for convolutional layers except for the last convolution layer which uses the sigmoid function. The fully connected layers are used to decrease (pooling) and increase (upsampling) the image size. As illustrated in Fig. 2, since the $N \times N$ input matrix is used to reconstruct a $M \times M$ image, the number of convolutional layers and fully connected layers will be asymmetrical in the encoder and decoder network.

As shown in Fig. 3(b)-(d), we have implemented three different types of EIT-CDAE models using different number of convolutional layers and fully connected layers. Our proposed EIT-CDAE architecture is scaling in the network because we can simply increase the complexity and depth of the networks with additional convolution layers and fully connected layers. However, the increase in the neural network size from model A to model C, will also result in an increase in the computational cost and memory. Therefore, in this work, we aim to evaluate the robustness of our proposed EIT-CDAE and compared it with the SAE architecture.

III. Experimental Results

A. *EIT database*

To evaluate the autoencoder models, we have chosen an open source database using a 16-electrode EIT system [17]. This EIT system was designed in 1994 by Ghislain Savoie and Robert Guardo and the recorded measurements are provided in the montreal database in 1995. We selected one of the experimental databases, zc_demo3, which is a series of measurement results based on a circular conductive rod in a 30 cm diameter acrylic tank. The entire dataset consists of 99 measurements. For a 16-electrode system, the measurement data is a 16×16 impedance information. We have used the EIDORS algorithm to reconstruct the image with a resolution of 64×64. We have used the impedance measurement data and EIDORS reconstructed images as the input and output of our network.

B. *Experimental Result*

In the experiments, we have implemented the SAE and EIT-CDAE models using the Tensorflow package with Python programming language [21]. These models have been evaluated using our 64-bit Linux workstation machine with an Intel i7-7800X 3.50GHz processor and an Nvidia GTX 2080 graphic card.

The first experiments are broken down into two parts. We have evaluated the quality of EIT reconstructed image against the three different types of EIT-CDAE models and experimented with the different types of loss function. The results of reconstructed EIT images are shown in the Fig. 4. We have manually annotated the images with blue circular markers, indicating the location of the rod.

From our observation in Fig. 4(a), the basic model of EIT-CDAE, model A, creates a coarse image quality with poorly defined boundary of the acrylic tank and unable to reconstruct the conductive rod. Both model B and C are able to reconstruct the EIT image with a homogeneous background without introducing noise artifacts. Among these models, model B achieved the best trade-off in terms of computational cost, memory usage, image quality, accuracy, speed and robustness.

To identify the most optimal loss function for our EIT-CDAE, we have evaluated the model B with poission, binary cross entropy and mean squared error functions. Fig. 4(b) has shown that the choice of loss function has a significant impact to the reconstructed images. Hence, we have adopted the binary cross entropy loss function to evaluate the SAE and our EIT-CDAE models.

C. Comparison with state-of-the-art works

To compare our reconstructed images with the state-of-the-art methods, we have implemented and evaluated the following image reconstruction algorithms: (1) EIDORS [7], [8] and (2) SAE [13]. As shown in the Fig. 5, the first, second and third row of images are EIT reconstructed images using the EIDORS, SAE and EIT-CDAE models with five different samples. The blue circle indicates the boundary of the circular conductive rod and the red outlines indicate the noise artifacts. Clearly, the reconstructed images using our EIT-CDAE have shown to have a better image quality and less noise artifacts.

The reconstructed images using the SAE model may easily introduce additional noise artifacts to the reconstructed images. We have also attempted to evaluate SAE models with different number of hidden layers and different number of nodes. However, the result seems to be consistent with the SAE's reconstructed images as shown in Fig. 5, Therefore, our proposed EIT-CDAE model has demonstrated to provide a higher image quality and reduces the noise artifacts.

IV. Conclusion

In this work, we have proposed and presented our preliminary EIT-CDAE model based on the autoencoder model with convolutional layers, to perform a 2-D EIT image reconstruction. The deep learning algorithm has been demonstrated to solve the ill-posed problem.

From our experimental data using phantom data, our EIT-CDAE model has reconstructed a better EIT image quality, removing any noise artifacts, making it more robust compared to the conventional stacked autoencoder and traditional non-linear algorithms. In our future work, we plan to further improve the algorithm to further enhance the image quality, relaxing the EIT hardware requirement.

References

[1] A. Adler and A. Boyle, "Electrical Impedance Tomography: Tissue Properties to Image Measures," *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 11, pp. 2494–2504, Nov 2017.

[2] R. P. Patterson, "Electrical Impedance Tomography: Methods, History, and Applications (Institute of Physics Medical Physics Series)," *Physics in Medicine and Biology*, vol. 50, no. 10, pp. 2427–2428, 2005.

[3] A. Adler, M. B. Amato *et al.*, "Whither lung EIT: Where Are we, Where do we Want to go and What do we Need to get There?" *Physiological Measurement*, vol. 33, no. 5, pp. 679–694, apr 2012.

[4] S. Hong, J. Lee *et al.*, "A 10.4 mW Electrical Impedance Tomography SoC for Portable Real-time Lung Ventilation Monitoring system," in *IEEE Asian Solid-State Circuits Conference*, Nov 2014, pp. 193–196.

[5] B. Liu, G. Wang *et al.*, "22.6 a 13-channel 1.53-mw 11.28-mm2 electrical impedance tomography soc based on frequency division multiplexing with 10 throughput reduction," in *IEEE International Solid-State Circuits Conference*, Feb 2019, pp. 370–372.

[6] Y. Gao, B. Liu *et al.*, "Live demonstration: A pulmonary conditions monitor based on electrical impedance tomography measurement," in *IEEE International Symposium on Circuits and Systems*, May 2019, pp. 1–1.

[7] A. Adler, J. H Arnold *et al.*, "Greit: A unified approach to 2d linear eit reconstruction of lung images," *Physiological measurement*, vol. 30, pp. S35–55, 07 2009.

[8] M. Vauhkonen, W. R. B. Lionheart *et al.*, "A matlab package for the eidors project to reconstruct two-dimensional eit images," *Physiological Measurement*, vol. 22, no. 1, pp. 107–111, 2001.

[9] G. Dong, R. Bayford *et al.*, "EIT Images with Improved Spatial Resolution Using a Realistic Head Model," in *International Conference of the IEEE Engineering in Medicine and Biology Society*, Aug 2006, pp. 1134–1137.

[10] B. Schullcke, S. Krueger Ziolek *et al.*, "Comparison of Image Reconstruction Algorithms in EIT Imaging," *Journal of Biomedical Science and Engineering*, vol. 09, pp. 137–142, 01 2016.

[11] S. Jalali and X. Yuan, "Using auto-encoders for solving ill-posed linear inverse problems," *CoRR*, vol. abs/1901.05045, 2019.

[12] M. T. McCann, K. H. Jin, and M. Unser, "Convolutional Neural Networks for Inverse Problems in Imaging: A Review," *IEEE Signal Processing Magazine*, vol. 34, no. 6, pp. 85–95, Nov 2017.

[13] X. Li, Y. Lu *et al.*, "An Image Reconstruction Framework Based on Deep Neural Network for Electrical Impedance Tomography," in *IEEE International Conference on Image Processing*, Sep. 2017, pp. 3585–3589.

[14] J. Adler and O. ktem, "Solving ill-posed inverse problems using iterative deep neural networks," *Inverse Problems*, vol. 33, no. 12, p. 124007, nov 2017.

[15] C. Tan, S. Lv *et al.*, "Image Reconstruction Based on Convolutional Neural Network for Electrical Resistance Tomography," *IEEE Sensors Journal*, vol. 19, no. 1, pp. 196–204, Jan 2019.

[16] S. J. Hamilton and A. Hauptmann, "Deep d-bar: Real-time electrical impedance tomography imaging with deep neural networks," *IEEE Transactions on Medical Imaging*, vol. 37, no. 10, pp. 2367–2377, Oct 2018.

[17] A. Adler and W. R B Lionheart, "Uses and abuses of eidors: an extensible software ease for eit," *Physiological measurement*, vol. 27, pp. S25–42, 06 2006.

[18] S. Salve, T. Shah *et al.*, "Automatization of Coloring Grayscale Images Using Convolutional Neural Network," in *International Conference on Inventive Communication and Computational Technologies*, April 2018, pp. 1171–1175.

[19] C. C. Tan and C. Eswaran, "Reconstruction and recognition of face and digit images using autoencoders," *Neural Computing and Applications*, vol. 19, no. 7, pp. 1069–1079, 2010.

[20] Z. Yanjun, W. Lili *et al.*, "Image reconstruction algorithm based on algebraic neural network for electrical resistance tomography," in *IITA International Conference on Control, Automation and Systems Engineering*, July 2009, pp. 242–245.

[21] M. Abadi, A. Agarwal *et al.*, "TensorFlow: Large-scale machine learning on heterogeneous systems," 2015, software available from tensorflow.org. [Online]. Available: https://www.tensorflow.org/